DriveCompose: Compositional Generative Reconstruction of Driving Scenes via Joint Completion

Supplementary Material

A. Cross-dataset generalization

We evaluate zero-shot transfer of Waymo-trained checkpoints on 150 nuScenes validation scenes. Each scene supplies one fixed, continuous 20-frame clip from FRONT, FRONT_LEFT, and FRONT_RIGHT. Native camera timestamps give a clip duration of approximately 1.67 s. Inputs at $\{0, 5, 10, 15\}$ leave 16 targets: 150 clips contain 1,800 context and 7,200 target images, including 5,400 interpolation and 1,800 extrapolation targets. Outputs are 240×160.

The supplied camera and actor states and shared state adapters match the Waymo conditions. All networks, codecs, and normalization parameters remain frozen, without target-domain training or scene adaptation. Table S1 reports means and sample SDs across training seeds. Ours improves over DGGT by 1.16 dB PSNR and 0.009 SSIM, supporting reconstruction transfer under these fixed conditions. Each dataset retains its native camera timing.

Table S1. Zero-shot reconstruction on nuScenes. Mean ± training-seed SD.

Method	PSNR ↑	SSIM ↑
STORM	17.77 ± 0.12	0.669 ± 0.003
DGGT	25.31 ± 0.08	0.794 ± 0.002
StreetForward	23.17 ± 0.10	0.712 ± 0.003
Ours	$\mathbf{26.47 \pm 0.07}$	$\mathbf{0.803 \pm 0.002}$

B. Training details

Learning is divided into four stages so that the completion target has a stable meaning. Recovery determines the evidence and support, the codec defines its latent coordinates, and the completion network learns a distribution over the missing codes. The last stage adapts this distribution to real observations without changing the representation it predicts.

Data and splits. Counts in Table S2 include only data passing quality checks. The original Waymo training set is divided into 740 training and 58 development sequences; the 58 development sequences select checkpoints, and the 202 validation sequences are used for testing. Synthetic maps, base vehicle models, and human identities are separated by source asset. Texture variants of a vehicle and clothing/motion variants of a human remain in one split. Real logs are deduplicated and separated by road segment, continuous capture, and identifiable actors. Completeness refers to geometry, materials, and motion bindings within the finite query support.

The synthetic test split supplies 100 scenes for paired hidden-surface evaluation and 20 scenes for asset editing. Fixed-asset reuse is also evaluated on 20 real scenes (Appendices H and I).

Complete targets on a fixed layout. Partial input establishes $\mathcal{Q}$. Both branches use the same query positions, scales, types, order, component ownership, and human bindings, following the block order and within-block indices of `QueryBundle.blocks`. Complete meshes are queried on this fixed support. For each non-sky query, the owning component's mesh is expressed in the query frame and intersected with its cell B_i. Positive-area triangle intersections supply surface targets. A cell with no intersecting surface receives a surface-free target after asset completeness checks.

For an intersecting triangle fragment f of area A_f, we draw

$$n_f = \max\left(1, \left\lceil \frac{64 A_f}{\delta_{\min}^2} \right\rceil\right) \tag{S1}$$

deterministic area samples, where $\delta_{\min}$ is the smallest cell side. Asset, query, and triangle indices determine sample locations. We render RGB, camera-axis depth, instance identity, and triangle identity using original materials and fixed illumination. Eight azimuths and three elevations form the base views. Local isolated views cover remaining fragments, providing features for surfaces occluded in the base views.

The Stage 1 visual encoder, frozen after recovery training, maps rendered RGB to 128-dimensional features. At each surface sample, we extract these features and append its normalized depth residual relative to the query center, clipped to $[-1, 1]$. We first average valid views of each sample with observation weights, then aggregate samples by fragment area to obtain $\mathbf{h}_i^{\text{full}} \in \mathbb{R}^{129}$. The partial branch retains query-center projection and visibility-weighted aggregation.

Surface-free cells use a 129-dimensional zero feature and explicit zero-opacity supervision. Surface-free targets describe the absence of mesh surface; free-space labels are supplied by observed rays. Sky targets come from the complete environment map with zero depth residual. Both branches share the visual feature basis, input format, and type-specific codec. The target code is

$$\mathbf{z}_{1,i} = E_{k_i}([\mathbf{h}_i^{\text{full}}; \mathbf{q}_i; \boldsymbol{\delta}_i]) \in \mathbb{R}^{128}. \tag{S2}$$

Table S2. Data and supervision. Scene maps and independent assets are split by source identity.

Data	Train	Dev.	Test	Supervision
Waymo sequences	740	58	202	RGB, calibration, ego poses, LiDAR, vehicle boxes and tracks
Synthetic scenes	10,000	1,000	1,000	Geometry, materials, depth, semantics, instances and actor states
Synthetic maps	80	10	10	Static background geometry and road structure
Vehicle assets	10,000	1,000	1,000	Canonical geometry, materials, dimensions and rigid frames
Human identities	2,500	250	250	Shape, texture, template, skeleton, skinning and joint states

The 135-dimensional input passes through a $135 \to 128 \to 128$ MLP and LayerNorm. Biases allow a zero feature to encode a nonzero latent. Hard projection separately enforces zero free-space codes during completion.

Stage 1: visual recovery. Real training logs and synthetic scenes supervise the image encoder, geometry heads, identity slots, and state heads. Real depth comes from training LiDAR, and vehicle states come from annotations. Confidence-weighted real semantic and human-joint auxiliary labels complement synthetic shape, skeleton, binding, and joint supervision. One trajectory-level matching assigns slots to identities and is reused across timestamps. Confidence is trained or calibrated against depth, occlusion, and reprojection reliability. Table S4 lists the loss weights.

Stage 2: latent representation fitting. With recovery frozen, we train the codec on synthetic scenes and complete assets, together with real RGB, depth, and silhouette observations. The objective is

$$\mathcal{L}_2 = \mathcal{L}_{\text{rgb}} + 0.1\mathcal{L}_{\text{depth}} + 0.2\mathcal{L}_{\text{silhouette}} + 0.05\mathcal{L}_{\text{empty}} + 0.001\mathcal{L}_{\text{scale}}. \quad \text{(S3)}$$

For rendered RGB $\widehat{\mathbf{I}}$ and depth $\widehat{d}$, the observation losses are masked L1 means,

$$\mathcal{L}_{\text{rgb}} = \langle \|\widehat{\mathbf{I}} - \mathbf{I}\|_1/3 \rangle_{\Omega_I}, \quad \text{(S4)}$$

$$\mathcal{L}_{\text{depth}} = \langle |\widehat{d} - d| \rangle_{\Omega_d}. \quad \text{(S5)}$$

Here $\langle \ell \rangle_\Omega$ averages valid elements. Surface-free cells receive squared-opacity loss $\mathcal{L}_{\text{empty}}$, alongside silhouette supervision and Gaussian-scale regularization. At the end of Stage 2, we freeze recovery and the codec and encode fixed complete targets $\mathbf{z}_1$ for prior learning.

Stage 3: completion-prior learning. The GCM learns the unknown-query flow-matching objective in Eq. (9) of the main paper. Every unknown query in an accepted synthetic sample has valid geometry, appearance, and binding targets. Samples with an invalid unknown target are excluded from this stage; the loss selector s_i operates within valid targets. Complete scenes provide cross-component context, and complete assets supply object geometry and appearance. Conditioning masks simulate contiguous unseen surfaces, background holes, long occlusions, short tracks, and uneven camera coverage. Pose, instance, and confidence perturbations expose the generator to recovery errors.

Stage 4: adaptation through rendering. We differentiate through flow integration, decoding, and rendering to update only the GCM. Complete synthetic samples and real logs from the 740 training sequences are mixed at 1:3. The objective is

$$\mathcal{L}_4 = \mathcal{L}_{\text{flow}} + \mathcal{L}_{\text{rgb}} + 0.1\mathcal{L}_{\text{depth}} + 0.05\mathcal{L}_{\text{free}} + 0.1\mathcal{L}_{\text{ground}} + 0.1\mathcal{L}_{\text{contact}} + 0.1\mathcal{L}_{\text{penetration}}. \quad \text{(S6)}$$

The flow term uses complete synthetic samples; each observation or geometry term uses samples carrying its required supervision. Every term is averaged over its own valid elements. With accumulated free-ray opacity a_r, foot signed distance d_f positive above ground, contacting feet Ω_c, and penetration depth d_p positive inside another actor, the geometry terms are

$$\mathcal{L}_{\text{free}} = \langle a_r^2 \rangle, \qquad \mathcal{L}_{\text{ground}} = \langle [-d_f]_+^2 \rangle,$$
$$\mathcal{L}_{\text{contact}} = \langle |d_f| \rangle_{\Omega_c}, \qquad \mathcal{L}_{\text{penetration}} = \langle [d_p]_+^2 \rangle, \quad \text{(S7)}$$

where $[x]_+ = \max(x, 0)$. Frozen decoding and hard code restoration preserve the observed anchors.

Optimization and inference. Table S5 gives the four-stage schedule. Training uses eight A100 80 GB GPUs, FP16, and AdamW with $\beta = (0.9, 0.95)$ and weight decay 0.01. Each stage uses warmup followed by cosine learning-rate decay. The 58 development sequences select checkpoints; the 202 validation sequences form the test set. The four stages total 287,360 updates and 1,001.5 GPU hours, including method-specific target preprocessing. Stage costs are 278.6, 184.3, 326.9, and 211.7 GPU hours. Three GCM training seeds share recovery and the codec, giving a combined cost of 2,078.7 GPU hours. The w/o CCM control reuses Stages 1–2 and retrains Stages 3–4 with the same budget. The deterministic w/o GCM control uses the fixed recovery module and codec.

Table S3. Network and support configuration. Observed anchors and canonical human-template queries augment the base grids.

Module	Configuration
Visual recovery	Width 128; patch size 16; 4 heads; 1 image-attention layer and 2 cross-view/time-summary layers
Identity and human state	24 identity slots; 10-dimensional shape code; 17-joint interface
Query features and latents	Feature dimension 129; latent dimension 128; type-specific $135 \to 128 \to 128$ encoders with LayerNorm
Completion network	Width 128; 4 layers; 4 heads; 16 learnable memory tokens
Gaussian decoding	4 children per query; 23 outputs per child, including 12 appearance parameters
Directional appearance	Constant RGB and three local directional coefficients per channel
Base supports	Background $20 \times 12 \times 6$ ($5 \times 5 \times 3$ m cells); rigid $8 \times 4 \times 4$ inside each asset box; sky 4×4 per face
Sampling	24 explicit Euler steps with observed-code restoration and free-space clearing after each step

Table S4. Stage 1 supervision weights.

Loss	Weight	Loss	Weight
Depth	1.0	Root rotation	0.2
Normals	0.2	Dimensions	0.1
Semantics	0.1	Human shape	0.1
Instances	0.1	Joints	0.2
Root translation	0.5	Confidence	0.1

At inference, the log is encoded once. Its query layout and observed codes remain fixed throughout 24 Euler steps, after which type-specific decoders produce assets for repeated state composition.

C. Scene composition details

State-driven rendering. At online step k, state s_k specifies the ego pose, active assets, rigid transforms, and human root and joint poses. A rigid asset's Gaussian center and covariance transform as

$$\boldsymbol{\mu}^w = \mathbf{R}\boldsymbol{\mu}^a + \mathbf{t}, \qquad \boldsymbol{\Sigma}^w = \mathbf{R}\boldsymbol{\Sigma}^a\mathbf{R}^\top. \tag{S8}$$

For humans, the weighted skinning transform is $\mathbf{A}_i = \sum_j w_{ij}\mathbf{P}_j\mathbf{B}_j^{-1}$, where $\mathbf{B}_j$ and $\mathbf{P}_j$ are rest and current global joints in the root frame. Weights satisfy $w_{ij} \geq 0$ and $\sum_j w_{ij} = 1$, giving $\mathbf{A}_i = \mathbf{I}$ in the rest state $\mathbf{P}_j = \mathbf{B}_j$. The transformed center is $\bar{\boldsymbol{\mu}}_i^w = \mathbf{T}_h^w\mathbf{A}_i\bar{\boldsymbol{\mu}}_i^h$. The combined local linear map $\mathbf{J}_i$ gives covariance $\mathbf{J}_i\boldsymbol{\Sigma}_i\mathbf{J}_i^\top$. Its inverse $\mathbf{J}_i^{-1}$ maps viewing directions into the canonical frame; the direction is normalized before appearance evaluation. The inverse is evaluated for nonsingular skinning transforms. The root pose controls overall placement, while the root joint retains its rest transform.

The renderer composites the background, active assets, and valid sky. External policy and dynamics can advance the state:

$$o_k = \mathcal{R}(\mathrm{Compose}(\mathcal{W}, s_k), \mathcal{C}_k), \tag{S9}$$

$$s_{k+1} = F_{\mathrm{dyn}}(s_k, \pi(o_k), \Delta t). \tag{S10}$$

Camera and actor state changes reuse the same background and canonical assets. Road meshes and collision proxies use the same coordinates as the visual components, allowing their motion to be driven by a physical simulator.

D. Visual recovery and aggregation details

Input conventions and preprocessing. The input is a synchronized log of images $\mathbf{I}_{t,v}$, timestamps, intrinsics $\mathbf{K}_v$, and camera-to-rig calibration $\mathbf{T}_{c_v}^r$. Rig poses $\mathbf{T}_r^w(t)$ are supplied in the controlled evaluations relative to the first frame, giving camera poses $\mathbf{T}_{c_v}^w(t) = \mathbf{T}_r^w(t)\mathbf{T}_{c_v}^r$. Transforms act on column vectors and distances are in meters. Images follow a rectified pinhole model and linear RGB; distortion, rolling shutter, and exposure differences are handled before evidence aggregation.

Shared visual encoding. The Driving Log Encoder uses patch tokenization [1] and transformer attention [3], with image-position, camera-calibration, and time embeddings. Within-image attention extracts spatial features. A second transformer processes pooled image summaries across views and timestamps, and returns the resulting context to the patch features. Geometry heads predict camera-axis depth, normals, confidence, semantics, and instance assignments. The confidence determines which predictions can contribute surface evidence.

Learned actor queries attend to all image tokens to form trajectory-level identity slots. Each slot then reads the features at an individual timestamp to estimate its state. This separates persistent identity, dimensions, and human shape from frame-dependent root poses and joint rotations. Root poses are predicted in the rig frame and mapped to world coordinates with $\mathbf{T}_r^w(t)$. A human shape code selects

Table S5. Four-stage training schedule. R: recovery; C: codec; G: GCM. D_R: real training logs; D_S: synthetic scenes; D_A: complete assets. Batch sizes are global.

Stage	Updated	Frozen	Data	Updates	Batch	Initial LR
Visual recovery	R	–	D_R, D_S	86,208	32	2×10^{-4}
Latent fitting	C	R	D_S, D_A, D_R	57,472	32	3×10^{-4}
Completion prior	G	R, C	Complete D_S, D_A	114,944	32	10^{-4}
Rendering adaptation	G	R, C	$D_S : D_R = 1 : 3$	28,736	8	3×10^{-5}

an identity-specific template with fixed canonical support, skeleton topology, and binding. Slot capacity bounds the number of represented actors.

Query supports. A query stores canonical position $\mathbf{q}_i$, cell scale $\boldsymbol{\delta}_i$, type k_i, local attention group g_i, and reference-world coordinate $\mathbf{q}_i^w$. Reliable depth samples provide surface anchors. Background anchors are combined with a finite world grid; rigid anchors are aligned by inverse SE(3) transforms and combined with a grid in the local bounding volume. Human queries lie on the canonical template surface. Sky queries are unit directions grouped by cubemap face. These supports provide explicit locations for both observed and missing content.

The layout remains fixed during completion. We obtain $\mathbf{q}_i^w$ by placing the query in the first input configuration, including skinning for humans. Canonical coordinates identify the content within its component; reference-world coordinates let the generator relate that component to its surroundings.

Metric support and anchor selection. The background base grid spans $[-20, 80] \times [-30, 30] \times [-3, 15]$ m with $20 \times 12 \times 6$ cells, giving $5 \times 5 \times 3$ m cell scales. Up to 2,048 observed anchors add 0.20 m support cells. Each vehicle uses an $8 \times 4 \times 4$ grid inside its actual bounding box, with cell sides $(L/8, W/4, H/4)$ and up to 512 observed anchors of side 0.20 m. Human queries lie on the canonical template with support neighborhoods of side 0.10 m. Sky has six cubemap faces with 4×4 direction queries per face. Background attention groups use 4 m spatial bins; this grouping scale is independent of the support-cell scales.

Valid estimated depths generate anchors in each branch's coordinates. We deduplicate by the bucket index $\lfloor \mathbf{p}/0.20 \rfloor$, retaining the first point in each bucket. If the unique-bucket count exceeds the branch cap, we select equally spaced indices in the ordered unique-bucket list. Anchor positions retain their sampled coordinates. Human queries carry template-provided binding weights before completion; all four decoded children inherit them. The root pose supplies global placement, and the internal root joint retains its rest transform.

Motion-aware evidence aggregation. For each observation, we place the canonical query at the corresponding actor state:

$$\mathbf{p}_i(t) = \begin{cases} \mathbf{q}_i, & k_i = \text{background}, \\ \mathrm{xyz}(\mathbf{T}_a^w(t)\bar{\mathbf{q}}_i), & k_i = \text{rigid}, \\ \mathrm{xyz}(\mathbf{T}_h^w(t)\mathbf{A}_i(t)\bar{\mathbf{q}}_i), & k_i = \text{human}. \end{cases} \tag{S11}$$

Here $\bar{\mathbf{q}}_i$ is homogeneous, xyz extracts Euclidean coordinates, and $\mathbf{A}_i(t)$ is the weighted skinning transform. Projection into a camera gives depth $z_{i,t,v}$ and sampled feature, depth, and confidence $(\boldsymbol{\phi}_{i,t,v}, d_{i,t,v}, c_{i,t,v})$. With tolerance $\eta = \eta_{\text{abs}} + \eta_{\text{rel}} d$, define $r = (d - z)/\eta$. A surface observation must lie inside the image, have valid depth and sufficient confidence, match the query's instance, and satisfy $|r| \le 1$. Background queries additionally require static-background semantics. Let $a_{i,t,v}$ denote this validity test. We aggregate

$$w_{i,t,v} = a_{i,t,v} c_{i,t,v} \exp(-r_{i,t,v}^2/2), \tag{S12}$$

$$\mathbf{h}_i = \frac{\sum_{t,v} w_{i,t,v} [\boldsymbol{\phi}_{i,t,v}; r_{i,t,v}]}{\max(\varepsilon, \sum_{t,v} w_{i,t,v})}. \tag{S13}$$

The query confidence is the maximum over accepted observations. The main reconstruction protocol uses center-ray rule C with $\eta = 0.15 + 0.02d$ meters and confidence at least 0.8. A reliable query-center ray votes for free space when

$$z_{i,t,v} + \tfrac{1}{2}\|\boldsymbol{\delta}_i\|_2 < d_{i,t,v} - \eta. \tag{S14}$$

Positive observation weight labels a query observed. Among the remaining queries, at least two free-space votes establish a free label; all others remain unknown. Sky projection uses camera rotation, reliable sky semantics, and zero depth residual, and receives no free-space votes.

Footprint rule and downstream reconstruction. We compare center-ray voting with a footprint rule on the same fixed query layouts in 100 synthetic test scenes. The footprint rule places all eight cell vertices using the query's rigid or skinning transform and computes

$$z_{i,t,v}^{\max} = \max_{x \in B_i(t)} z_v(x), \qquad \Omega_{i,t,v} = \pi_v(B_i(t)). \tag{S15}$$

The projected footprint is rounded outward to cover every intersected depth pixel. A view votes only when the footprint lies fully inside the image, all depths are valid, every confidence is at least 0.8, and sky and ego-vehicle pixels are absent. It must satisfy

$$z^{\max}_{i,t,v} < \min_{p \in \Omega_{i,t,v}} \left[D_{t,v}(p) - 0.15 - 0.02 D_{t,v}(p) \right]. \quad \text{(S16)}$$

Other views abstain. The two-vote rule, observed-surface priority, and hard clearing are retained.

The footprint rule reduces occupied-cell errors from 1,374 to 19 (Table S6). With reference depth, center-ray voting still mislabels 934 occupied cells, localizing an error source to lateral cell coverage. Continuous reference-depth footprint bounds yield zero such errors on this audit set. The remaining 19 errors with discrete estimated depth arise mainly at subpixel structures and depth boundaries.

From mask labels to reconstructed surfaces. The two rules start from the same intermediate state before the first hard clearing. We fix weights, query layout, input depth, postprocessing, and per-query random noise. On the same 100-scene hidden-region evaluation, F improves PSNR by 0.18 dB, coverage by 1.2 percentage points, and F-score by 1.3 points (Table S7). Thin structures, occlusion boundaries, and human surfaces gain 2.7, 2.2, and 1.0 points of hidden-surface coverage. Human semantic slices can overlap the structural slices and are scored separately. All coverage entries here use a 2 cm point-to-mesh threshold; human asset editing uses 1 cm.

The mask audit counts 99,184 reference-surface-containing query cells. Its disjoint categories describe cell-label errors; hidden-surface coverage instead measures reference-surface recall using area-uniform samples. The 1,355 net reduction in mislabeled cells (98.62%) and the 1.2-point coverage gain measure these different stages. Ray violations use the 2,020,000 fixed-ray denominator in Appendix J.

On Waymo input views, F gives 30.406 dB PSNR and 0.1215 m depth MAE, versus 30.391 dB and 0.1220 m for C. Added occlusion changes from 0.18% to 0.17%, and strict free-ray violations from 0.07% to 0.08%. Both rules preserve observed Gaussian parameters exactly. Construction takes 2.86 s with F versus 2.66 s with C, a 7.5% increase on the RTX 4090. Appendix J reports the rendering measurements, and Appendix K specifies timing conditions.

E. Gaussian decoding details

A Gaussian stores its center $\boldsymbol{\mu}$, covariance $\boldsymbol{\Sigma}$, opacity α, and directional appearance.

Background, rigid, and human Gaussian decoders independently map each query to $K = 4$ child Gaussians. Each child has 23 output parameters: three center offsets, three scales, four rotation parameters, one opacity, and twelve appearance coefficients. For child k of query i, the center is confined to its support cell:

$$\boldsymbol{\mu}_{i,k} = \mathbf{q}_i + \tfrac{1}{2}\boldsymbol{\delta}_i \odot \tanh(\mathbf{b}_{i,k}). \quad \text{(S17)}$$

Positive scales and normalized rotations determine positive-definite covariance. Opacity is multiplied by $1 - f_i$, so free queries cannot produce visible Gaussians. Human children inherit the skinning weights of their support query; rigid children share their asset's rigid transform. A separate sky head predicts cubemap colors, with the observation-supported domain retained in V_{sky}.

Appearance uses a constant RGB term and three local directional coefficients per channel, evaluated as $c(d) = \max(0, c_0 + C_{\text{app}}\hat{d}_{\text{local}})$. Decoding is per query: once the codec is frozen, an unchanged observed code produces unchanged Gaussian parameters. The completion network can exchange context across queries without changing this decoding property.

F. Reconstruction protocols and baselines

The main comparison reconstructs scenes under supplied camera and actor states. The last four frames evaluate conditional temporal extrapolation. Table S10 lists test conditions. Target human joints come from independent offline annotation/fitting, and human shape and binding are recovered from context. Test LiDAR is reserved for evaluation. All test-time network weights remain fixed.

Training resources. We retrain STORM, DGGT, and StreetForward from random initialization, preserving their native architectures. Every method uses the same real training logs D_R, synthetic scenes D_S, and complete assets D_A. Baselines receive RGB, depth, silhouette, and complete-asset target views through their native objectives. Ours additionally uses cell-empty and template-binding supervision. Checkpoints are selected on the shared development set within each fixed training budget (Table S8). Matching recovery, codec, data, and optimization in the CCM ablation isolates the contribution of component exchange.

Shared state adapter. The baselines use a common deterministic state adapter. Visible Gaussian rendering contributions are projected into the same context instance masks and accumulate weighted identity votes. Vehicles are canonicalized with the inverse supplied root SE(3), transforming centers, covariances, and directional coordinate bases together. Human Gaussians obtain 17-joint skinning weights by barycentric interpolation on corresponding triangles of the context-recovered identity template, followed by inverse skinning. The adapter preserves predicted appearance and adds no prediction network. Unbound outputs remain

Table S6. Free-space mask audit on 100 synthetic scenes. Counts measure cells containing reference surfaces that receive a free label. Categories are mutually exclusive. C denotes center-ray voting and F the projected-footprint rule. Main reconstruction tables use C; F is evaluated separately.

Category	Surface-containing cells	Center-ray errors	Footprint errors
Thin structures	18,437	614	9
Occlusion boundaries	27,619	587	7
Other regions	53,128	173	3
Total	99,184	1,374	19
Error / surface-containing cells		1.385%	0.019%
Number of free labels		47,286	32,871
Surface contamination / free labels		2.906%	0.058%

Table S7. **Downstream effects of free-space voting.** Shared checkpoints and per-query noise, with nine seed combinations. Coverage and F-score use 2 cm.

Metric	C	F
Hidden PSNR ↑	23.18	**23.36**
Hidden LPIPS ↓	0.192	**0.185**
Hidden coverage ↑	95.0%	**96.2%**
Hidden F-score ↑	88.6%	**89.9%**
Thin-structure coverage ↑	92.1%	**94.8%**
Boundary coverage ↑	93.0%	**95.2%**
Human coverage ↑	95.8%	**96.8%**

scored, and actor edits that cannot be assembled count as failures.

For the same Gaussian set, inverse and forward placement produce a maximum linear-RGB difference below 4×10^{-6}. Converting a native time-varying representation into fixed canonical assets changes cross-time reconstruction as shown in Table S9. The adapted column is used in the main comparison.

The first 200 valid synchronized frames per test sequence form ten non-overlapping 20-frame clips, totaling 2,020 clips. Context indices are $\{0, 5, 10, 15\}$ for three cameras. Twelve interpolation frames yield 72,720 images, and the four extrapolation frames $\{16, 17, 18, 19\}$ yield 24,240, totaling 96,960 targets. Every target timestamp has valid dynamic masks and depth-evaluation samples. Aggregating Table S11 at a 3:1 interpolation/extrapolation ratio recovers the main-table means. Depth scale alignment is used only for evaluation, with no feedback to reconstruction or rendering.

Depth scale and metric evaluation. The aligned depth columns use a positive per-frame scale on a fixed valid LiDAR mask M_t,

$$s_t = \text{median}_{p \in M_t} \frac{d_t(p)}{\widehat{d}_t(p)}. \tag{S18}$$

Only scale is fitted. Table S12 uses the same masks and frame–clip–sequence aggregation for aligned and unaligned depth. Alignment is applied inside the depth evaluator; reconstruction, composition, surface distances, and RGB reprojection use the original metric coordinates.

G. Ablation protocols and seed statistics

The two learned variants use training seeds $\{0, 1, 2\}$ and generation seeds $\{7, 17, 27\}$ for each checkpoint, giving nine combinations. We first average generation seeds, then frames, windows, sequences, and training seeds equally, without best-of-three selection. In Tables S13 and S14, each entry is the mean ± sample SD of the three training-seed means; parentheses give the average sample SD across generation seeds. Zero variation for w/o GCM describes deterministic inference with shared, fixed recovery and codec.

Each of the 202 test sequences contributes one 29-timestamp window. Inputs $\mathcal{T}_2 = \{0, 28\}$, $\mathcal{T}_4 = \{0, 8, 20, 28\}$, and $\mathcal{T}_8 = \{0, 4, \ldots, 28\}$ share seven target timestamps $\{2, 6, \ldots, 26\}$ and three cameras, giving 4,242 target images per combination. Two learned variants, three input settings, and nine seed combinations yield 54 groups and 229,068 predictions. The deterministic w/o GCM control adds 4,242, totaling 233,310.

Non-renderable counts are $0/10,908$ learned-variant window instances and $0/202$ for w/o GCM. Low quality, holes, and contact errors remain scored. All variants share preprocessing, 240×160 resolution, annotation-based dynamic masks, and equal sequence weighting. Training samples $T \in \{2, 4, 8\}$ uniformly, and inference uses 24 Euler steps. Hyperparameters are selected only on the development split.

At $T = 2, 4, 8$, CCM increases full-image PSNR by 1.104, 0.738, and 0.627 dB and reduces LPIPS by 0.041, 0.028, and 0.018 (Table S14). The largest gains occur with the sparsest input.

Table S8. Training resources for one complete checkpoint, including method-specific target preprocessing. Ours and w/o CCM include the shared recovery and codec stages.

Method	Training objective	Updates	Global batch	GPU hours
STORM	Native reconstruction plus same-source synthetic views	234,816	16	916.7
DGGT	Native geometry/rendering plus same-source synthetic views	212,672	16	1,128.4
StreetForward	Native scene reconstruction plus same-source synthetic views	246,144	8	1,036.9
Ours	Four-stage training	287,360	By stage	1,001.5
w/o CCM	Shared R/C; retrained GCM prior and adaptation	287,360	As Ours	977.1

Table S9. Effect of conversion to fixed canonical assets. PSNR is in dB.

Method	Native	Adapted	Change
STORM	26.492	26.380	−0.112
DGGT	27.538	27.410	−0.128
StreetForward	27.094	27.010	−0.084

H. Off-trajectory rendering and completion analysis

Each offset direction renders $2,020 \times 3 \times 20 = 121,200$ images and 6,060 videos. Videos contain 20 frames at 10 Hz and never concatenate clips. The reference population consists of original-trajectory recordings from the same test clips, cameras, and timestamps. FID uses a fixed Inception-v3 with 2,048-dimensional features; FVD uses a fixed I3D with 1,024-dimensional features. Left and right scores are computed separately and averaged, without pooling the two populations first. GT in the main off-trajectory figure denotes the original-trajectory image.

One hundred synthetic test scenes separately provide paired evaluation of regions never observed in the inputs (Table S15), using representative shifts of ± 2 m. FID/FVD describe distribution quality, whereas hidden-region metrics evaluate scene-corresponding completion.

Predicted surfaces and geometry metrics. We construct surfaces by rendering and fusion. Gaussians with opacity at least 0.01 are evaluated within 3σ. Six axis-aligned orthographic views are rendered in blocks over a common evaluation domain, with pixel size at most 1 mm inside each block. A ray's first accumulated-opacity crossing of 0.5 defines a surface depth; rays below this opacity have no predicted surface. We fuse depths into a sparse TSDF with 2 mm voxels and an 8 mm truncation distance, and extract the mesh in valid, supported voxels, leaving unknown regions open. Native neural fields use the same accumulated-opacity threshold, fusion, and distance evaluation.

Predicted and reference meshes each supply 65,536 area-uniform samples. A BVH computes each sample's nearest distance to the other triangle mesh. Coordinates are in meters and distances are non-squared Euclidean distances. Evaluation keeps the original metric scale and alignment. With predicted samples P, reference samples G, and corresponding surfaces S_{pred} and S_{GT} in the evaluation region,

$$P_\tau = \frac{1}{|P|} \sum_{p \in P} \mathbf{1}[d(p, S_{\text{GT}}) < \tau], \tag{S19}$$

$$R_\tau = \frac{1}{|G|} \sum_{g \in G} \mathbf{1}[d(g, S_{\text{pred}}) < \tau], \tag{S20}$$

$$F_\tau = \frac{2 P_\tau R_\tau}{P_\tau + R_\tau}. \tag{S21}$$

Coverage is reference-surface recall R_τ. We compute metrics per scene and seed before averaging. Asset coverage uses only surfaces assigned to that actor, with 2 cm vehicle and 1 cm human thresholds. Complete reference geometry and input visibility define newly visible surfaces; reference surfaces outside query support remain in the coverage denominator.

Real-view correspondences. Independent reference geometry establishes correspondences between recorded and off-trajectory views. Dynamic actors are transformed with shared states. The reference evaluator removes occlusion-ambiguous correspondences for every method; finite outputs and rendering coverage determine method validity. Test LiDAR is used only by the evaluator. Linear-RGB MAE is measured on both each method's valid subset and their intersection, which covers 58.7% of the reference set (Table S16). CCM reduces common-support MAE from 0.028 to 0.019 and raises overall coverage from 86.9% to 92.4% relative to w/o CCM.

Task-adapted completion mechanisms. The scene-latent control adapts the scene-level Gaussian latent generation mechanism of CompleteSplat [2]; the object-coupling

Table S10. Test conditions shared by DriveCompose, STORM, DGGT, and StreetForward.

Condition	Shared setting
Context RGB	The same four timestamps and three cameras
Camera and ego state	Shared intrinsics, extrinsics, and ego poses
Test depth/LiDAR	Not an input; depth is estimated from RGB
Identity, boxes, root tracks	Shared supplied states
Human shape and binding	Recovered from context; common binding interface
Target human joints	Shared independent offline annotations/fits
Test-scene optimization	None for any method

Table S11. Temporal interpolation and conditional extrapolation. Image columns report PSNR/SSIM; depth reports scale-aligned AbsRel/δ_1.

Method	Time group	Full image	Dynamic region	Depth
STORM	Interp.	26.800/0.838	22.600/0.640	0.170/0.753
STORM	Extrap.	25.120/0.790	20.600/0.576	0.202/0.705
DGGT	Interp.	27.800/0.806	23.200/0.667	0.237/0.680
DGGT	Extrap.	26.240/0.758	21.240/0.603	0.285/0.632
StreetForward	Interp.	27.400/0.830	21.500/0.608	0.185/0.771
StreetForward	Extrap.	25.840/0.782	19.176/0.544	0.233/0.723
Ours	Interp.	**28.900/0.909**	**25.500/0.870**	**0.091/0.885**
Ours	Extrap.	**27.556/0.877**	**23.700/0.838**	**0.107/0.853**

Table S12. Depth evaluation with and without per-frame scale alignment.

	Per-frame scale aligned		Unaligned	
Method	AbsRel ↓	δ_1 ↑	AbsRel ↓	δ_1 ↑
STORM	0.178	0.741	0.226	0.683
DGGT	0.249	0.668	0.327	0.572
StreetForward	0.197	0.759	0.254	0.694
Ours	**0.095**	**0.877**	**0.121**	**0.842**

control adapts JRM's cross-object latent coupling [4]. Both operate through the shared recovery module, codec, query layout, and state interface. They replace the completion interaction mechanism while using the same input evidence, training data, supervision, and 24-step sampling setup. The shared recovery and codec stay fixed. The completion training schedule is given in Appendix B, and comparison scores average three training and three generation seeds without sample selection.

Table S15 uses the same 100 synthetic test scenes and 12,000 shifted target images per seed combination. Each learned variant produces 108,000 outputs across nine seed combinations; scene-wise scores include all views, frames, and seeds. Over the object-coupling control, CCM adds 0.67 dB PSNR, reduces LPIPS by 0.020, and increases coverage and F-score by 2.6 and 3.6 percentage points. The PSNR gain over the scene-latent control is 0.94 dB. Surface evaluation uses original metric coordinates, 65,536 area-uniform samples from each mesh, and a shared 2 cm point-to-triangle threshold.

Visibility-stratified synthetic evaluation. Table S17 reuses the same 100 synthetic test scenes, ± 2 m shifts, inputs, checkpoints, and generated samples as the paired hidden-region evaluation. Each scene supplies one 20-frame clip with three cameras and context timestamps $\{0, 5, 10, 15\}$ (12 images). Targets cover both shift directions, all 20 timestamps, and three cameras, giving 12,000 images per seed combination. Context timestamps are also valid targets because the shifted cameras differ from the input cameras. Three training seeds and three generation seeds reuse trained weights without adapting on these scenes.

Visibility v is the fraction of a fixed reference surface region's area covered by the union of input views. Well-observed regions have coverage $\geq 80\%$, severely missing regions $\leq 20\%$, and intermediate regions $20\% < v < 80\%$. Reference-surface correspondences assign target pixels to background, vehicle, or human and to a visibility group. Methods use the same valid masks; group-wise frame means are averaged with equal scene weight. Group PSNRs and full-image PSNR are computed independently. The hidden-region test requires exactly 0% input coverage, whereas the severely missing group permits up to 20%.

CCM adds 1.02, 1.28, and 1.54 dB for severely missing

Table S13. Full seed statistics for component ablations.

Metric	w/o GCM	w/o CCM	Ours
Full-image PSNR	27.137 ± 0	28.286 ± 0.074 (0.032)	**28.913 ± 0.061 (0.028)**
Full-image SSIM	0.867 ± 0	0.893 ± 0.001 (0.0006)	**0.911 ± 0.001 (0.0005)**
Full-image LPIPS	0.183 ± 0	0.147 ± 0.002 (0.0013)	**0.129 ± 0.002 (0.0011)**
Dynamic PSNR	23.431 ± 0	24.683 ± 0.112 (0.048)	**25.617 ± 0.095 (0.041)**
Dynamic SSIM	0.811 ± 0	0.849 ± 0.002 (0.0010)	**0.873 ± 0.002 (0.0008)**

Table S14. Input-timestamp sensitivity with fixed checkpoints; $T = 8$ reuses the component results.

T	Variant	PSNR ↑	SSIM ↑	LPIPS ↓
2	w/o CCM	25.237 ± 0.103 (0.055)	0.817 ± 0.003 (0.0011)	0.254 ± 0.004 (0.0023)
2	Ours	**26.341 ± 0.089 (0.047)**	**0.851 ± 0.002 (0.0009)**	**0.213 ± 0.003 (0.0019)**
4	w/o CCM	27.126 ± 0.086 (0.041)	0.869 ± 0.002 (0.0008)	0.187 ± 0.003 (0.0018)
4	Ours	**27.864 ± 0.072 (0.035)**	**0.891 ± 0.001 (0.0006)**	**0.159 ± 0.002 (0.0014)**
8	w/o CCM	28.286 ± 0.074 (0.032)	0.893 ± 0.001 (0.0006)	0.147 ± 0.002 (0.0013)
8	Ours	**28.913 ± 0.061 (0.028)**	**0.911 ± 0.001 (0.0005)**	**0.129 ± 0.002 (0.0011)**

Table S15. Paired evaluation of synthetic hidden regions.

Variant	PSNR	LPIPS	Coverage	F-score@2 cm
w/o GCM	16.82	0.401	61.5%	51.7%
w/o CCM	21.56	0.236	89.7%	81.3%
Scene latent	22.24	0.221	91.8%	84.1%
Object coupling	22.51	0.212	92.4%	85.0%
Ours	**23.18**	**0.192**	**95.0%**	**88.6%**

Table S16. Linear-RGB reprojection error and coverage. The common support covers 58.7% of the reference set.

Variant	Common MAE	Own MAE	Coverage
w/o GCM	0.022	0.024	62.1%
w/o CCM	0.028	0.041	86.9%
Ours	**0.019**	0.029	**92.4%**

background, vehicle, and human regions (Table S17). Well-observed regions gain 0.04, 0.04, and 0.03 dB, concentrating the benefit where input coverage is limited.

Read-scope interventions on Waymo. Table S18 uses the same 202 Waymo windows as the main $T = 8$ ablation. Each 29-timestamp window uses $\{0, 4, 8, 12, 16, 20, 24, 28\}$ from three cameras (24 inputs), with targets $\{2, 6, 10, 14, 18, 22, 26\}$ (4,242 images per seed combination). Training seeds $\{0, 1, 2\}$ and generation seeds $\{7, 17, 27\}$ give nine combinations. Recovery, codec, query layout, target states, and 24-step sampling are fixed.

Reading only background, other vehicles, or other humans applies inference-time masks to the same Ours checkpoints, without separate retraining. Local attention and the global log condition remain unchanged. The w/o CCM endpoint uses the per-component-memory control retrained with the same budget. Thus 28.286 and 28.913 dB reuse the main ablation endpoints, while 28.721, 28.574, and 28.352 dB measure read interventions under that Waymo protocol. Waymo read interventions and synthetic visibility scores are reported separately; component gains overlap.

Table S17. PSNR stratified by component and input visibility.

Component and visibility	w/o CCM	Ours	Gain
Background, well observed	30.20	30.24	+0.04 dB
Background, severely missing	22.10	23.12	**+1.02 dB**
Vehicle, well observed	28.30	28.34	+0.04 dB
Vehicle, severely missing	21.10	22.38	**+1.28 dB**
Human, well observed	26.90	26.93	+0.03 dB
Human, severely missing	19.70	21.24	**+1.54 dB**

Table S18. CCM read-scope diagnostics.

Extra readable components	Full-image PSNR
No exchange (w/o CCM)	28.286
Background	28.721
Other vehicles	28.574
Other humans	28.352
All components	**28.913**

I. State-driven asset reuse

Twenty real and twenty identity-disjoint synthetic test scenes each contain one tested vehicle and human. We evaluate the

Table S19. Structural integrity of real actor edits with training seed 0 and generation seed 7. Columns give successes.

Variant	Vehicle	Human	Overall
w/o GCM	14/20	10/20	60.0%
w/o CCM	18/20	15/20	82.5%
Ours	**20/20**	**17/20**	**92.5%**

Table S20. Structural-integrity successes across training and generation seeds. Each row evaluates twenty vehicles and twenty humans.

Training	Generation	Vehicle	Human	Total
0	7	20	17	37/40
0	17	20	18	38/40
0	27	19	17	36/40
1	7	20	18	38/40
1	17	19	17	36/40
1	27	20	18	38/40
2	7	20	17	37/40
2	17	20	18	38/40
2	27	19	17	36/40
All combinations		177/180	157/180	334/360

original state, a 1 m lateral camera shift, a 30° vehicle rotation about its root and the world vertical axis, and a human pose change at fixed root placement. Every operation reuses the same completed asset and generation seed under fixed lighting, with appearance held fixed. Complete reference geometry defines newly visible surfaces as described in Appendix H.

Real-scene edit ratings. Training seed 0 and generation seed 7 give forty edits (Table S19). Two evaluators blinded to method identity independently inspect predefined target views; a third adjudicates disagreements. Structural integrity fails when a continuous hole, detached surface, or joint tear spans at least five pixels at the evaluation resolution. Ours succeeds in all twenty vehicle rotations and seventeen human edits. Its three human failures involve a large elbow bend, leg self-occlusion, and loose-clothing boundaries. Vehicle dimensions and human shape parameters remain fixed, and previously observed human texture has canonical linear-RGB MAE 0.014.

Seed variation and scene clustering. The three training seeds $\{0, 1, 2\}$ and three generation seeds $\{7, 17, 27\}$ yield 360 edits on the same twenty real scenes (Table S20). Vehicles succeed in 177/180 edits (98.33%), humans in 157/180 (87.22%), and the aggregate in 334/360 (92.78%). The sample SD of the three training-seed mean structural-integrity rates is 0.48 percentage points. Scene-cluster bootstrap sampling retains all eighteen edits for a selected scene, giving a 95% structural-integrity interval of 85.28–99.17%. The 26 failures are concentrated in four scenes with 9, 8, 6, and 3 failures.

Newly visible texture, identity, and geometry. An asset surface is newly visible when it is invisible in every input observation and visible after editing. Canonical surface correspondences and visibility tests define these regions. Five independent raters inspect newly visible texture, appearance identity, and local geometry; each dimension passes with at least four positive judgments. Texture ratings inspect blur, repetition, stretching, and seams. Identity ratings inspect vehicle shape and paint, and human clothing, body shape, and visible identity features. Geometry ratings inspect wheel/body connections, limb volume, joint bends, and intersections.

Table S21 scores 360 actor edits from twenty scenes and nine seed combinations; camera shifts add no actor-edit trials. Joint acceptance is the per-edit intersection of the four criteria, yielding 298 passes (82.78%). The 62 failures comprise 26 structural failures and 36 structurally intact edits that fail at least one other quality check.

Extended state and view protocol. The 360-edit quality evaluation uses fixed target-state and view lists. The expanded protocol applies vehicle yaw changes of 30°, 60°, and 90°, and human maximum non-root joint rotation differences of 15°, 30°, and 45° at fixed root placement. Twenty scenes and nine seed combinations yield $20 \times 9 \times (3+3) = 1,080$ actor edits. Three predefined target views per edit give 3,240 images. Each seed reuses one completed asset across all states, with the five-pixel criterion defined at 240×160 resolution.

J. Observation consistency

We fix original cameras, actor states, and evaluation pixels, using the observed-only scene as the pre-completion reference (Table S22). Observed Gaussian parameters are preserved exactly, while added Gaussians can change rendering through occlusion and alpha compositing. We fix 10,000 rays per sequence, totaling 2,020,000. Added occlusion means added content produces accumulated opacity above 0.01 more than 5 cm in front of an observed surface. Strict free-ray violation requires opacity above 0.05 more than 10 cm in front of the surface.

For Ours, 0.18% added occlusion corresponds to 3,636 rays and 0.07% strict violation to 1,414. Relative to the pre-completion scene, input-view PSNR decreases by 0.029 dB and depth MAE increases by 2 mm. Removing the free-ray loss raises the two ray rates to 1.34% and 0.91%, showing

Table S21. **Real actor-edit quality under rule C.** Twenty scenes, nine training/generation-seed combinations, and 360 actor edits. Joint acceptance is the per-edit intersection.

Criterion	Vehicle (180 edits)	Human (180 edits)	All (360 edits)
Structural integrity	177 (98.33%)	157 (87.22%)	334 (92.78%)
New-surface texture	166 (92.22%)	154 (85.56%)	320 (88.89%)
Appearance identity	174 (96.67%)	170 (94.44%)	344 (95.56%)
Local geometry	164 (91.11%)	152 (84.44%)	316 (87.78%)
All four criteria	156 (86.67%)	142 (78.89%)	298 (82.78%)

Table S22. Input-observation preservation under C. Depth MAE is in meters. The matched C/F comparison is reported in Appendix D.

Configuration	Parameter Δ	PSNR	RGB MAE	Depth MAE	Occlusion	Violation
Before completion	0	30.420	0.0187	0.120 m	0	0
w/o CCM	0	30.368	0.0190	0.123 m	0.24%	0.09%
Ours	**0**	**30.391**	**0.0189**	**0.122 m**	**0.18%**	**0.07%**
Ours w/o free-ray loss	0	29.970	0.0208	0.134 m	1.34%	0.91%

its effect on rendering beyond query-level parameter preservation.

K. Efficiency and reuse cost

Training uses $8\times$ A100 80 GB GPUs. Inference timing uses one RTX 4090 24 GB, FP16, batch 1, and 240×160 output, with an average of 1.4 million non-transparent Gaussians per scene. After 20 warmups, 100 repetitions include CPU–GPU transfers and synchronization but exclude disk reads and initial model loading. Construction takes 2.66 ± 0.11 s, state composition plus one camera render takes 7.8 ± 0.4 ms (approximately 128.2 FPS), and peak memory is 14.2 GB.

Let $Q = Q_b + \sum_a Q_a^{\text{vehicle}} + \sum_h Q_h^{\text{human}}$ count non-sky queries. Decoding produces $4Q$ candidate Gaussians, which bound the non-transparent output count. The mean of 1.4 million therefore corresponds to at least 350,000 non-sky queries per scene on average. Sky uses the separate cubemap head.

Table S23 breaks down construction. The w/o CCM control takes 2.53 s, so CCM adds 0.13 s (approximately 5.1%). Table S24 computes total time including one construction as $2.66 + 0.0078N$. Here N counts single-camera outputs; a three-camera query counts as three renders.

Changing only the free-space rule to F raises construction time from 2.66 to 2.86 s (7.5%) under the same hardware, input, and sampling conditions. The main timing breakdown and amortized costs use C.

Table S23. Scene-construction timing breakdown.

Stage	Time
Image preprocessing	0.18 s
Visual recovery	0.64 s
Queries and canonical aggregation	0.17 s
24-step completion	1.38 s
Gaussian decoding	0.26 s
Initial scene registration	0.03 s
Total	**2.66 s**

Table S24. Amortized query costs for fixed completed assets.

N	Total time	Per query
1	2.6678 s	2.6678 s
10	2.738 s	273.8 ms
100	3.440 s	34.4 ms
1,000	10.460 s	10.46 ms

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